

Do Touchdown Passes Predict Passing Yards?

NFL 2024–25 Season · n = 32 QBs · OLS Regression · Control: Attempts

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Uncontrolled Model

$$\text{Yards} = 1010.17 + 99.14(\text{TD})$$

Variable	Coef.	Sig.
Intercept	1010.17	p < 0.001
TD coef.	99.14	p < 0.001
R ²	0.790	
F-stat	112.7	p < 0.001

Controlled Model (+Attempts)

$$\text{Yards} = 138.15 + 47.58(\text{TD}) + 4.55(\text{Att})$$

Variable	Coef.	Sig.
Intercept	138.15	p = 0.436
TD coef.	47.58	p < 0.001
Att coef.	4.55	p < 0.001
R ²	0.927	
F-stat	184.1	p < 0.001

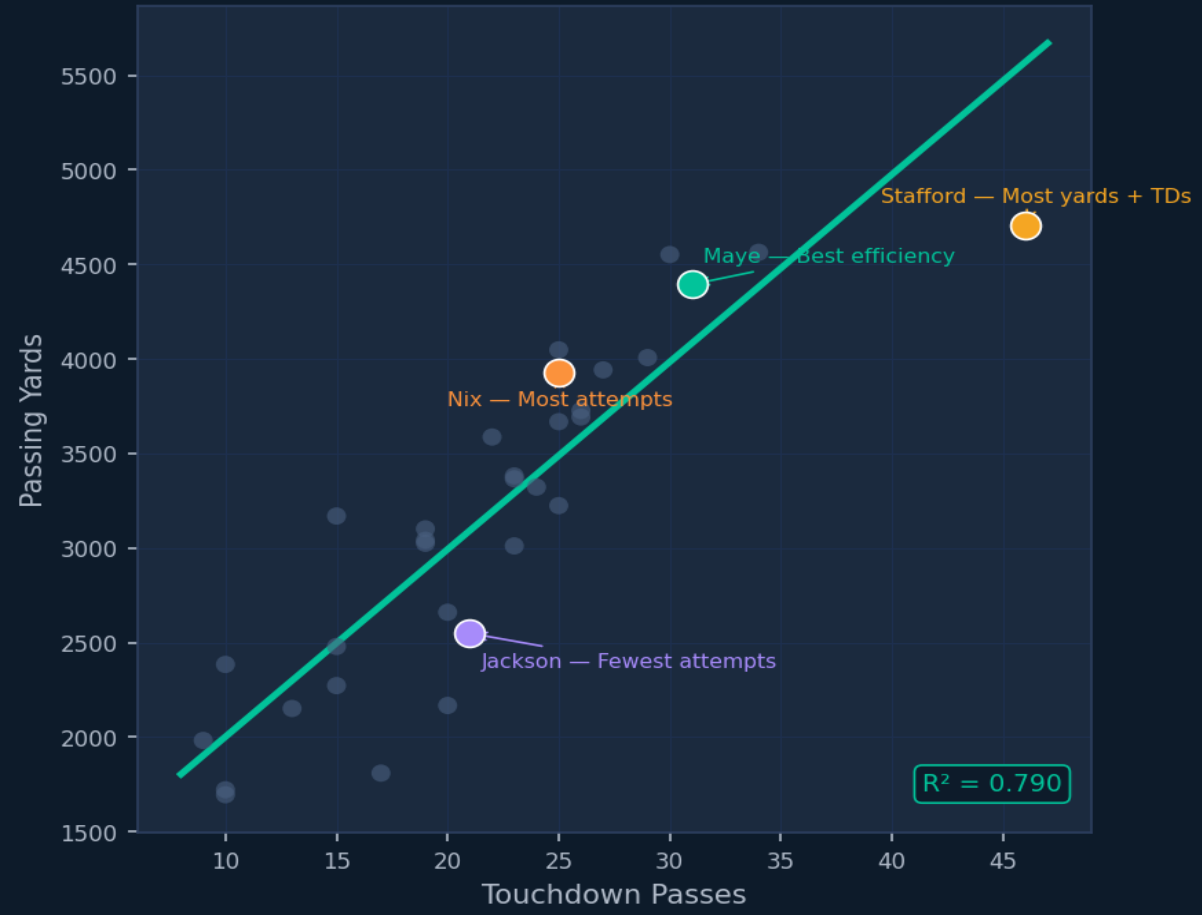
Controlling for Attempts halves the TD coefficient (99 → 48), revealing that throwing volume inflates the uncontrolled estimate. TDs remain highly significant.

Key Findings

NFL 2024–25 · OLS Regression

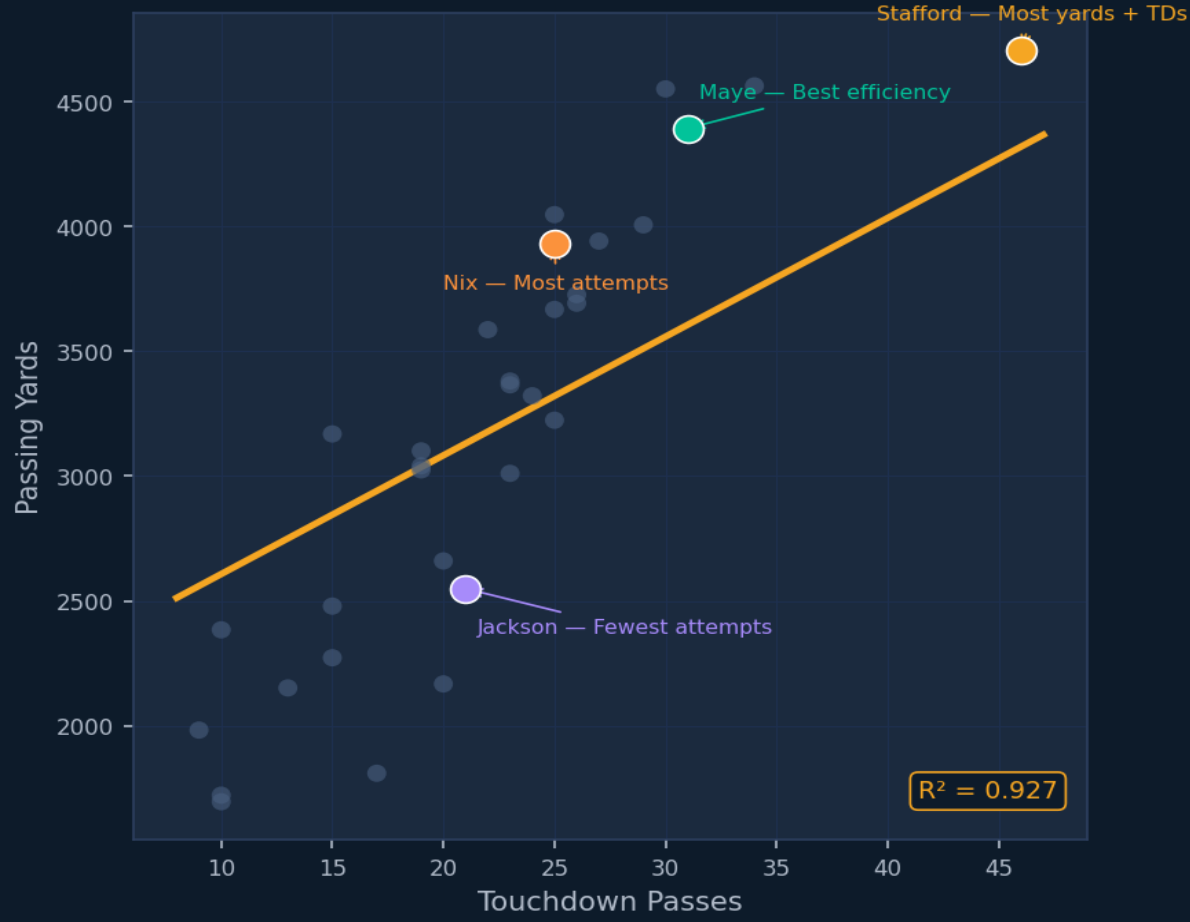
Uncontrolled Model

$$\text{Yards} = 1010 + 99.1(\text{TD})$$



Controlled Model (Att held at mean = 438)

$$\text{Yards} = 138 + 47.6(\text{TD}) + 4.55(\text{Att})$$



TD slope drops 99.1 → 47.6 after controlling for Attempts | volume was inflating the uncontrolled model

* Each additional TD predicts ~48 more yards holding volume constant ($p < 0.001$). Controlling for Attempts raises R^2 from 0.790 → 0.927, showing volume was inflating the uncontrolled TD effect.